

FOUND

Assignment

ABC : Data Case Study

Introduction

- As a member of ABC's Data team, you've been tasked with analyzing and predicting potential risks that our company may face. Your work will be crucial in helping us make informed decisions to mitigate those risks and ensure the continued success of our business.
- We've provided you with a sample dataset that includes information about our customers and their transaction history. Your task is to analyze the data and provide insights on any potential risks that we may face.
- This interview is designed to help us learn about your analytical capabilities, problem-solving abilities, and communication skills.

Problem

- Time estimate: 2-3 hours
- Provide a 1-page brief on what rule(s) should be implemented to prevent future "bad" businesses using historical information included in the data sample. Include any supporting files-notebooks, google sheets, etc.--as well.

Key areas to cover

- Description of the rule(s)
- Rationale for the rule(s) based on provided data
- Effectiveness of the rule(s)

Data Sample

- businesses.csv

- business_token: unique identifier for a business on ABC
- is_bad: boolean to indicate whether we have determined this business is fraudulent
- transactions.csv
 - business_token: unique identifier for a business on ABC
 - transaction_date: timestamp of when the transaction occurred
 - entry_type: type of transaction
 - atm: use of the ABC debit card at an ATM
 - ach_internal: ACH transfer initiated by ABC
 - ach_external: ACH transfer initiated by another financial institution
 - card: use of the ABC debit card
 - check: a check deposit
 - linkfound_to_found_transfer: a transfer between two ABC accounts
 - instant_debit: funding of a ABC account via an external debit card
 - direction: Indicating the flow of funds; either to a ABC account or from a ABC account
 - amount_cents: positive amount of a transaction in USD cents
 - transaction_status: final status of a transaction
 - internal_failure_reason: enumeration for why a card transaction was declined
 - vendor_name: name of other party in a transaction

— the following is only for card transactions

- mcc: merchant category code for a card transaction
- mcc_description: Friendly name for a merchant category code
- merchant_country
- pos_data_type: Type of point of sale
- token_wallet_type: Present if was tokenized transaction (i.e. Apple Pay)

- pos_data_attended: Indicates if the point-of-sale was attended at time of transaction
- pos_data_on_premise: Indicates if the point-of-sale was on-premise
- card_entry_mode: indicates if card was or was not present during the transaction
- pan_entry_mode: Indicates how the card information was entered
- pin_entered: indicates if PIN was entered during the transaction
- card_network
- avs_result: indicates the status of the address verification

Think big, start small

To help describe what we are looking for in this problem, here is the description of one of ABC's operating principles:

At the end of the day, we are here to build something of lasting impact. We make ambitious plans and prioritize quick progress. We only learn how our product resonates once it is in the hands of our customers. Sometimes we will be right in our hypothesis, sometimes, we will learn we were wrong, but we don't know either way until we get feedback. With this in mind, we prioritize shipping over waiting for perfection. We seek the highest impact and next step forward while keeping our sights set on our long-term vision. We default to moving things forward rather than waiting. We articulate a plan and let others edit/push back rather than wait to take the first step.

Submission

Success

Your assignment was submitted!

